Mohammad Asif Hasan

DC Road, Chawkbazar, Chittagong, Bangladesh +880-1827027919
asif78h78@gmail.com linkedin.com/in/asif-hasan-001b36212
github.com/Asif190 kaggle.com/mohammadasifhasan

Summary

- B.Sc. graduate in Electrical and Electronic Engineering from BUET, working as an **AI System Development Engineer** at BRIM AI, building production-grade AI agents for real-time security surveillance applications.
- Experienced in designing, training, and deploying deep learning models across computer vision, multimodal AI, NLP, and edge inference pipelines — with hands-on deployment on resource-constrained hardware (Raspberry Pi 4, mobile platforms).
- 4 peer-reviewed conference publications, 1 arXiv preprint, and multiple top IEEE competition placements including **IEEE VIP CUP 2025 Second Runner Up**; proven ability to deliver high-performance AI systems under real-world constraints.

Education

Bangladesh University of Engineering and Technology (BUET) — **Dhaka, Bangladesh**
B.Sc. in Electrical and Electronic Engineering (EEE) — *Jan. 2022 – Jul. 2026*
CGPA: 3.19/4.00

Notre Dame College — **Dhaka, Bangladesh**
Higher Secondary School Certificate (HSC), Science; GPA: 5.00/5.00 — *2018 – 2020*

Nasirabad Govt. High School — **Chittagong, Bangladesh**
Secondary School Certificate (SSC), Science; GPA: 5.00/5.00 — *2016 – 2018*

Experience

BRIM AI — **Present**
AI System Development Engineer — *Dhaka, Bangladesh*
- Developing an AI agent for intelligent room security surveillance, integrating computer vision and real-time threat detection pipelines.
- Building and deploying the model into a user-facing mobile application, ensuring optimized inference and seamless user experience.

Bangladesh Data Center Company Limited (BDCCL) — **Nov. 2025**
Industrial Attachment – National Data Center, Bangabandhu Hi-Tech City — *Gazipur, Bangladesh*
- Visited and studied the operations of a Tier-IV certified national data center covering power infrastructure (UPS, VRLA battery banks, 2.5 MW diesel generators), cooling systems (CRAC/CRAH, chiller plants, hot/cold aisle containment), fibre network backbone, and cloud service provisioning.
- Observed NOC/SOC operations, VESDA-based fire detection, NAFS-124 inert gas suppression, and multi-layered physical & cybersecurity architecture.

Thesis

Undergraduate Thesis — EEE 400, BUET — **Jan. 2026 – Jun. 2026**
Supervisor: Dr. Lutfa Akter, Professor, Dept. of EEE, BUET — *Dhaka, Bangladesh*
- **Title:** *Toward Clinically Interpretable Breast Cancer Imaging: A Boundary-Aware Hybrid Framework for Ultrasound Lesion Segmentation and Multimodal Mammographic Classification with BI-RADS-Aware Fusion*
- **Part I:** Designed a Lesion-Aware Attention U-Net with a novel composite loss and introduced the **Lesion Preservation Rate (LPR)** metric — first in pectoral segmentation literature. Achieved DSC = 0.897, LPR = 97.7% (+16.6 pp over best baseline, $p<0.001$).
- **Part II:** Built a hybrid attention-based segmentation framework (residual encoding, windowed self-attention, boundary-aware loss) validated on BUSI (77.28%), UDIAT (82.17%), and BUSBRA (88.61%) — outperforming five baselines under rigorous cross-validation.
- **Part III:** Developed **BOCAF** (Bidirectional Ordinal Cross-Attention Fusion) for multimodal BI-RADS classification, fusing mammogram features with clinical metadata. Achieved QWK = 0.793, AUC = 0.841 on VinDr-Mammo.
- **Datasets:** CBIS-DDSM, VinDr-Mammo, BUSI, UDIAT, BUSBRA **Tools:** PyTorch, NumPy, scikit-learn, OpenCV, LaTeX

Publications

Peer-Reviewed Conference Proceedings

[1] **M. A. Hasan**, N. Ahmed, and M. Ahmed, "Real-time Breast Cancer Classification Employing GMM-HMRF Segmentation and PCA-Optimized Machine Learning Algorithms," in *2025 28th International Conference on Computer and Information Technology (ICCIT)*, Cox's Bazar, Bangladesh, 2025. [IEEE Xplore]

[2] **M. A. Hasan**, M. S. Hasan, A. Hamim, N. Ahmed, S. Imam, and T. Ahmed, "Multimodal Pipeline for Sentiment Mining via MAG-BERT, LoRA-T5 Generated Suggestions with SHAP Explanations for Curriculum Optimization," in *2025 28th International Conference on Computer and Information Technology (ICCIT)*, Cox's Bazar, Bangladesh, 2025. [IEEE Xplore]

[3] F. Labiba, **M. A. Hasan**, S. Sarkar, A. Shahriar, N. Tasnim, and S. A. Fattah, "Real-Time Multi-Modal Drone and Bird Tracking Using Modality-Aware Adaptation and Physics-Aware Filtering," in *2025 IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE)*, Cox's Bazar, Bangladesh, 2025. [IEEE Xplore]

[4] A. S. M. Asaduzzaman, M. M. Hasan, M. K. Jahan, M. A. A. Chowdhury, **M. A. Hasan**, and M. Arman, "An Improved Optimization Approach for Secure User Authentication in Cyber Physical Systems," in *International Conference on Computing Technologies*, Bengalore, India, 2025.

Preprints

[5] S. Sarkar, **M. A. Hasan**, K. A. Shahriar, F. Labiba, N. Tasnim, and S. A. H. Fattah, "EGD-YOLO: A Lightweight Multimodal Framework for Robust Drone-Bird Discrimination via Ghost-Enhanced YOLOv8n and EMA Attention under Adverse Condition," *arXiv preprint arXiv:2510.10765*, 2025. [arXiv]

Projects

HIVE-R: Bio-Inspired Swarm Drone System for Human Detection in GPS-Denied Rescue Environments | GitHub

- Edge AI Deployment: Optimized and deployed a 6.2 MB YOLOv8n model on Raspberry Pi 4 achieving 86.9% mAP under adverse smoke/dust conditions with 185 ms inference latency — production-level edge AI engineering.
- System Integration: Co-developed a "Queen-Worker" multi-UAV swarm framework with EKF-fused monocular VO and IMU for 6-DOF navigation, integrating perception, planning, and communication into a unified autonomous pipeline.
- Real-Time Telemetry: Built a 915 MHz MAVLink pipeline transmitting 3D coordinates to GCS with $<$500 ms end-to-end latency, enabling real-time operator situational awareness.

Adaptive Multimodal Sentiment Analysis and Feedback Summarization | GitHub

- Engineered an end-to-end multimodal AI pipeline combining MAG-BERT (cross-modal fusion), LoRA-T5 (parameter-efficient summarization), and SHAP (explainability) for automated feedback analysis and curriculum optimization.
- Applied LoRA fine-tuning to reduce trainable parameters while maintaining generation quality — a key technique for deploying LLMs in resource-constrained production environments.

Breast Cancer Detection via Neural Network & Image Segmentation (MATLAB) | GitHub

- Built a full computer vision pipeline implementing GMM-HMRF probabilistic segmentation followed by PCA-optimized ML classifiers for real-time automated breast cancer detection.
- Designed modular preprocessing and classification components mirroring production ML pipeline best practices: data ingestion, feature extraction, model inference, and output postprocessing.

Variational Quantum Classifier for Iris Classification: A NISQ-Ready Implementation | GitHub

- Implemented a VQC using Qiskit 1.0+, comparing `TwoLocal` and `EfficientSU2` ansatz architectures with COBYLA optimization, achieving 65–80% test accuracy on a quantum simulator.
- Benchmarked model robustness under simulated NISQ noise (depolarizing + readout errors), gaining practical intuition for near-term quantum hardware constraints.

Quantum Simulation of "Particle in a Box" Using Schrödinger Equation in MATLAB | GitHub

- Numerically solved the time-independent Schrödinger equation via finite-difference discretization, constructing the Hamiltonian matrix and computing eigenvalues/eigenvectors for quantized energy levels and wavefunction visualization.

Automatic Vending Machine (IC-Based Digital System) | Demo

- Designed and implemented a complete FSM-based automated billing system using combinational and sequential digital logic ICs.

Competitions & Awards

IEEE VIP CUP 2025 **2025**

Second Runner Up *ICIP 2025*

- Developed a custom YOLOv8 architecture for accurate detection and high-FPS inference; created a model to detect payload in drones using RGB and IR modalities.

Datathon | CUET CSE Fest 2025 **2025**

Finalist

- Developed a novel pipeline using Paddle-OCR to classify memes as political or non-political.

NNO Content Writing Hunt Programme

Top 5

Divisional Science Fair 2017 **2017**

2nd Position

Affiliation & Volunteering

BUET Robotics Society **Mar. 2025 – Jul. 2026**

Vice President *BUET, Dhaka, Bangladesh*

IEEE BUET EMBC SB Chapter, Bangladesh **Jun. 2025 – Present**

Webmaster *BUET, Dhaka, Bangladesh*

IEEE **Jan. 2021 – Present**

Student Member

Newspaper Olympiad™ **Jul. 2021 – May 2022**

Content Lead

Grants

ICIP 2025 Travel Grant **2025**

IEEE VIP CUP 2025 Competition

Technical Skills

ML Frameworks & Libraries: TensorFlow, Keras, PyTorch, NumPy, Pandas, Matplotlib, scikit-learn

LLM & AI Agent Tools: LlamaIndex, FastAPI, Qiskit 1.0+

Programming Languages: Python, C, C++, HTML, CSS, JavaScript, Verilog

DevOps & Cloud: Docker, Git, GCP (Google Cloud Platform), Linux

Dev Tools & Utilities: MATLAB, Jupyter Notebook, Anaconda, LaTeX, Microsoft Office

Circuit Simulation & Design: PSpice, PSAF, Proteus, Arduino

Relevant Coursework

- Artificial Intelligence and Machine Learning
- Digital Signal Processing
- Introduction to Medical Imaging
- Introduction to Robotics
- Communication Systems
- Linear Algebra
- Probability & Statistics

Other Experiences

Conducted a session on Intro to Robotics and Arduino Programming – January 2025 [Post]
Bangladesh University of Engineering and Technology, Dhaka, Bangladesh